FULL STACK DEVELOPMENT WITH MERN

PROJECT DOCUMENTATION

Online Complaint Registration System

Prepared By: THUM SUCHITH REDDY

Project Documentation for Academic Submission

1. Introduction

The Online Complaint Registration System is a web-based platform developed using the MERN Stack. It provides a centralized mechanism for users to register complaints, track their progress, and receive updates. The system reduces paperwork, increases transparency, and improves complaint resolution efficiency.

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2. Project Overview

The project aims to digitize complaint management. Users can submit complaints online while administrators manage and resolve them through a dashboard. The platform supports authentication, complaint tracking, notifications, and reporting.

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3. Problem Statement

Traditional complaint handling methods rely on paper forms, emails, and manual processes. These approaches lead to delays, poor tracking, and lack of accountability. A centralized digital system is required.

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4. Objectives

• Digitize complaint management
• Improve transparency
• Enable complaint tracking
• Reduce manual effort
• Improve communication
• Increase user satisfaction

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5. System Architecture

The system follows a three-tier architecture:
Presentation Layer (React.js)
Business Logic Layer (Node.js + Express.js)
Database Layer (MongoDB Atlas)

Flow: User → Frontend → Backend APIs → Database → Response

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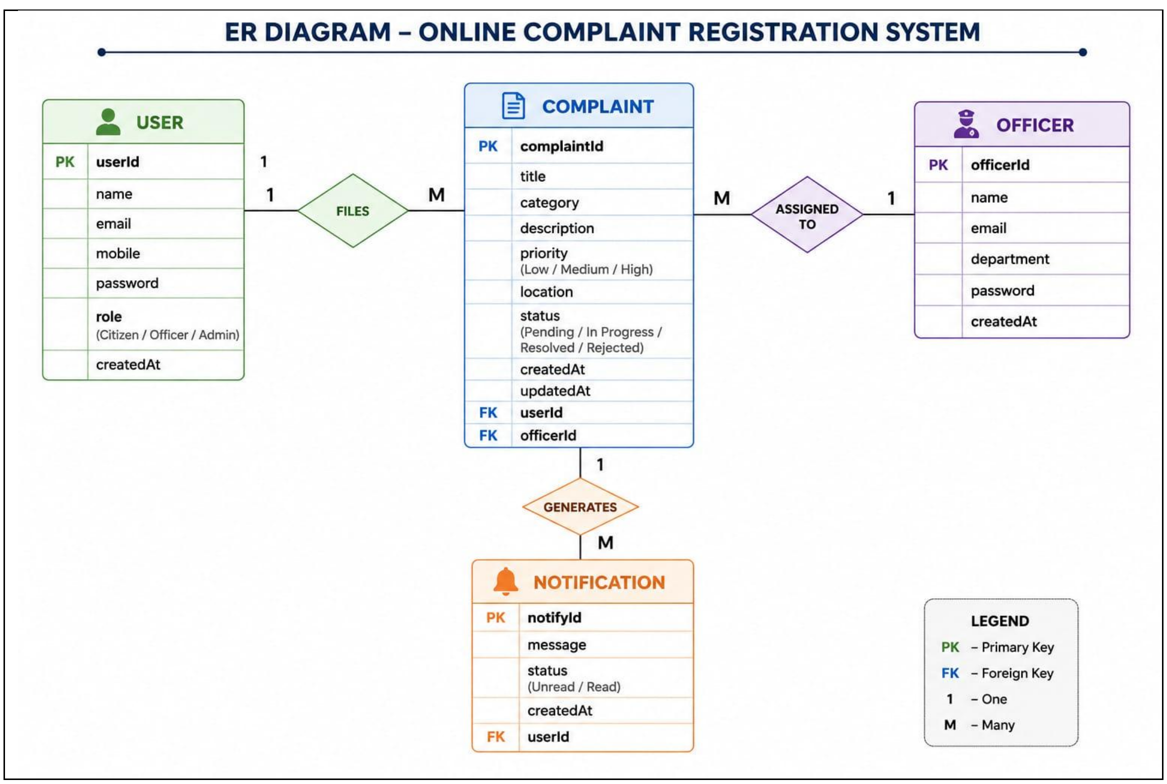

6. Technology Stack

Frontend: React.js, Vite, Axios, React Router DOM
Backend: Node.js, Express.js
Database: MongoDB Atlas, Mongoose
Authentication: JWT, bcryptjs
Deployment: Vercel, Render
Version Control: Git, GitHub

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7. Setup Instructions

Install Node.js, Git, and MongoDB Atlas account.
Clone repository.
Install dependencies using npm install.
Configure .env variables.
Run frontend and backend servers.

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8. Folder Structure

Client: components, pages, services, routes
Server: config, controllers, middleware, models, routes
This structure ensures maintainability and scalability.

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9. Application Workflow

User Registration → Login → Submit Complaint → Complaint Review → Status Update → Resolution → Notification

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10. API Documentation

Authentication APIs:
POST /register
POST /login

Complaint APIs:
POST /complaints
GET /complaints
PUT /complaints/:id

Admin APIs:

GET /all-complaints
PUT /status-update

Detailed Discussion: Authentication APIs:
POST /register
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11. Authentication & Security

JWT Authentication, bcrypt password hashing, protected routes, role-based access control, secure API communication.

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12. User Interface

The application includes Login Page, Registration Page, User Dashboard, Complaint Submission Form, Complaint Tracking Page, Admin Dashboard, and Notification Center.

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13. Testing

Functional Testing, API Testing, Integration Testing, and User Acceptance Testing were performed. All major modules passed successfully.

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14. Deployment

Frontend: https://vip-c2-online-complaint-registerati.vercel.app/
Backend: https://complaint-registeration.onrender.com
Database: MongoDB Atlas

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15. Known Issues

Email notifications and SMS integration are not yet implemented. Mobile optimization can be improved further.

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16. Future Enhancements

AI-based complaint categorization, analytics dashboard, SMS alerts, mobile application, multilingual support, and automated complaint routing.

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17. Conclusion

The Online Complaint Registration System successfully demonstrates the implementation of a full-stack MERN application. It improves transparency, efficiency, and user satisfaction while providing a scalable architecture for future growth.

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Live Project Links

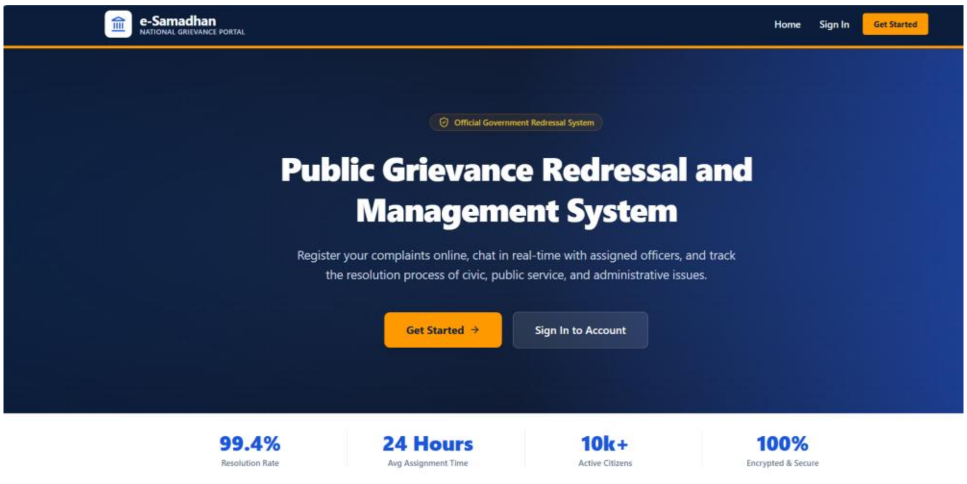
Frontend: https://vip-c2-online-complaint-registerati.vercel.app/

Backend: https://complaint-registeration.onrender.com

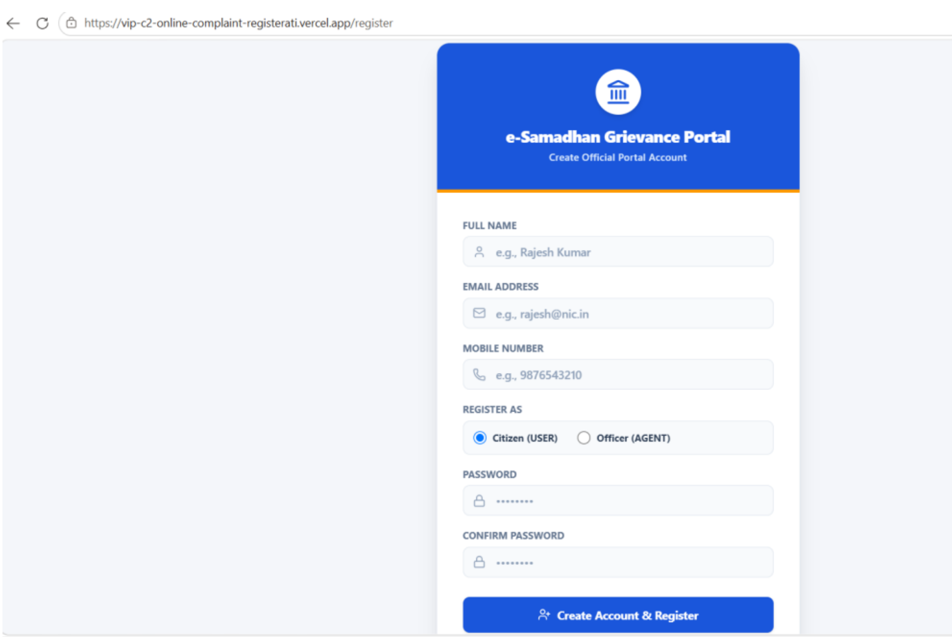
GitHub: https://github.com/suchithreddy-07/-VIP-C2--Online-complaint-registration

ONLINE COMPLAINT REGISTRATION SYSTEM

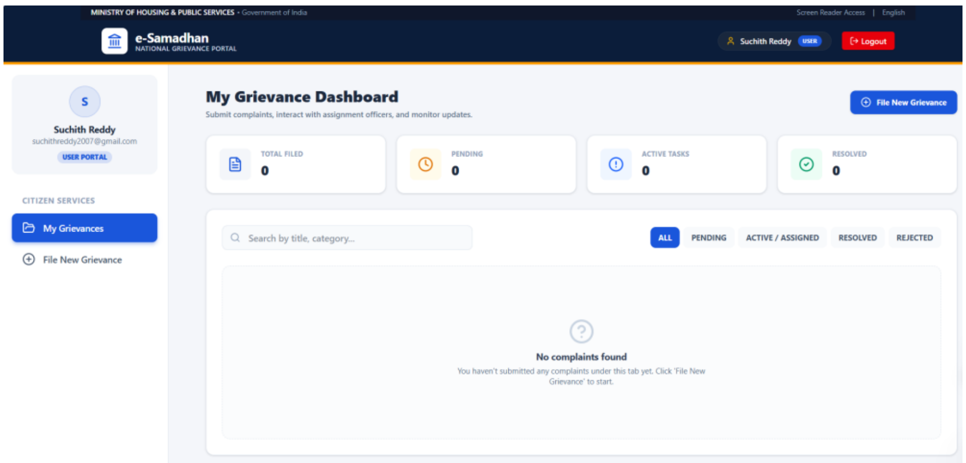
Home Page



Registration Page



User Dashboard



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